# Aluvextra Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-313/A

## Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. The control strategy links critical quality attributes to critical process parameters identified during development. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Deviations observed during the campaign were investigated and closed with no impact to product quality. Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

## Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. The control strategy links critical quality attributes to critical process parameters identified during development. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

### Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

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## Control of Excipients (3.2.P.4)

For the commercial formulation F-313/A, Kollidon CL (Crospovidone, disintegrant) is used.

Grade selection and control follow the principles of FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

### Excipient Grade Selection

Kollidon CL was selected as the commercial grade of crospovidone on the basis of lower moisture content mitigating hydrolytic degradation of the drug substance.

### Registered Specification

The current approved commercial specification (SPEC-EX-5893 Rev 08) lists Crospovidone as Kollidon CL.

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Cleaning verification swab results were below the health-based exposure limits derived for the product. No changes to the approved analytical procedures are proposed in this submission. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

## Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. No changes to the approved analytical procedures are proposed in this submission. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

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## Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

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